

COMPUTER NETWORKING

PROJECT NAME: CAMPUS NETWORK



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1. INTRODUCTION:

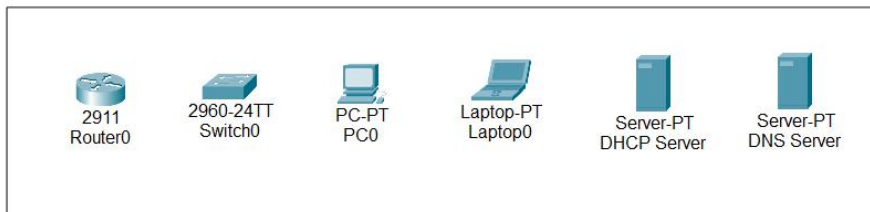
1.1 PROJECT STATEMENT:

In this mini-project, we defined a simulation of campus networks based on bus topology network. The network is divided into 4 departments: Software engineering, Computer science, physics department, IT department and also contain lab, admin, and library. The major aim of this project is to show the wire connectivity that is used in universities to make the network efficient and mobile at the same time. Mobility is the major concentration of this project. In order to provide equal functionality to all the users (college staff and students), we have added DNS and DHCP servers for the maximum utilization of resources. Hence the campus network provides different services such as connecting the user to the internet, data sharing among users (students, teachers, and different university members), accessing different web services for different functionalities, so it needs wired networking for smooth processing.

2. NETWORK REQUIREMENTS:

2.1 Devices Used In The Network:

Devices	Quantity
1) Router (1941)	8
2) Switches (2960-24TT)	8
3) DHCP server	1
4) DNS server	1
7) PCs	22
8) Laptops	4



Devices used in a network

2.2 Implementation:

- To design the campus network of the university we initially started by placing the core devices into the frame as mentioned in the above.
- Firstly, we placed the routers in a straight line in order to achieve the bus topology of the university outline, which was further connected to the switches using the gigabit ethernet port with copper straight-through cable and connect PCs, laptops and server to those switches.
- Router 1 connected to the switch 1 which further connects to the PCs of SE and CS department making vlan 10 and vlan 20.
- The DHCP router0 connected to the switch0 and that was further connected to the 5 PCs (IT department)
- Router 2 connected to the switch 2 which further connects to DHCP server and 1 PCs.
- Router 3 connected to the switch 3 which further connected to DNS server and PCs.
- Router 4 connected to the switch 4 making VPN tunnel to the Router7 which further connects to switch 7 of physics department.
- Similarly, Router 5 and Router 6 connected to Switch 5 and switch 6 respectively, further connects to PCs.
- All the routers connected to each other through copper cross-over wire.

2.3 Use of Bus topology:

- Bus topology is used to connect devices in a linear sequence, making it easy to install and maintain.
- It's often used in small networks, such as in a single room or building.

- **Advantages:**

1. **Easy installation:** Devices can be added or removed easily without disrupting the entire network.
2. **Cost-effective:** Requires less cable than other topologies, making it a cost-effective option.
3. **Simple maintenance:** Faulty devices can be easily identified and replaced.
4. **Scalability:** Can be expanded by adding more devices to the bus.

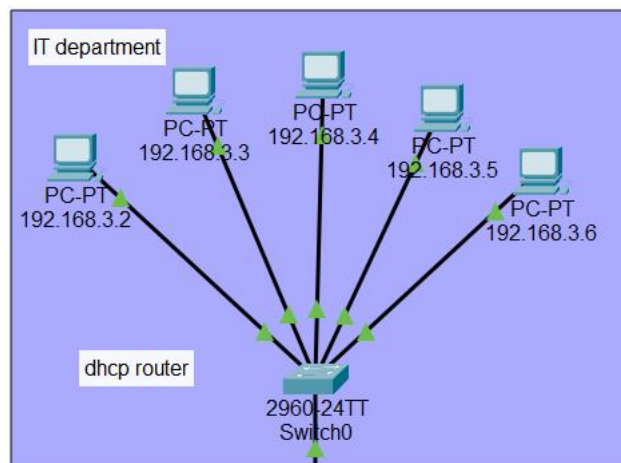
- **Disadvantages:**

1. **Single point of failure:** If the backbone cable fails, the entire network goes down.
2. **Limited size:** Bus topology is not suitable for large networks due to signal attenuation and noise.
3. **Difficult troubleshooting:** Can be challenging to identify the source of problems due to the shared cable.
4. **Performance issues:** As more devices are added, network performance can degrade due to increased traffic and collisions.
5. **Limited security:** Since all devices share the same cable, security risks are higher if a device is compromised.

3. CONFIGURING IP ADDRESSING:

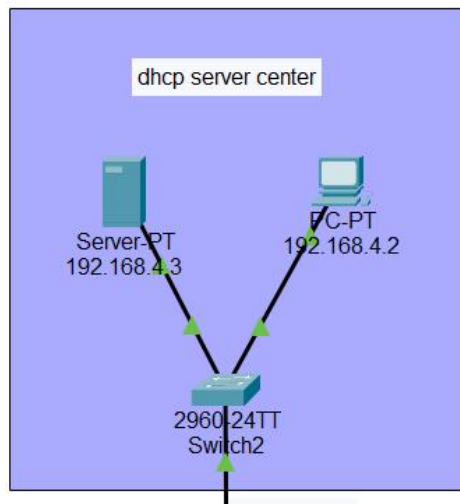
We have attached the screenshots of all the IP configuration below:

3.1 IT Department (DHCP Routing):

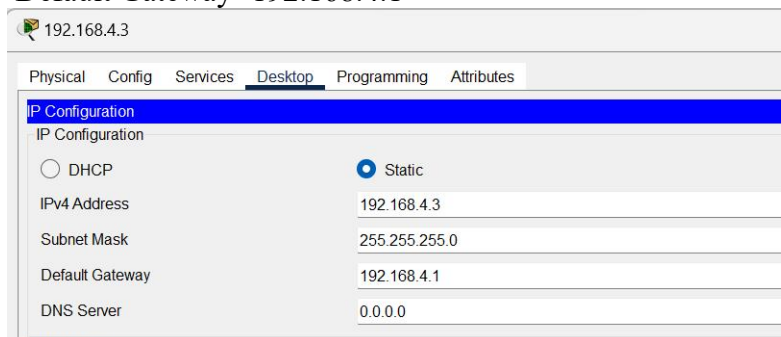


IP Addresses are as follows
192.168.3.2- PC1
192.168.3.3- PC2
192.168.3.4- PC3
192.168.3.5- PC4
192.168.3.6- PC5
Subnet Mask- 255.255.255.0
Default Gateway- 192.168.3.1

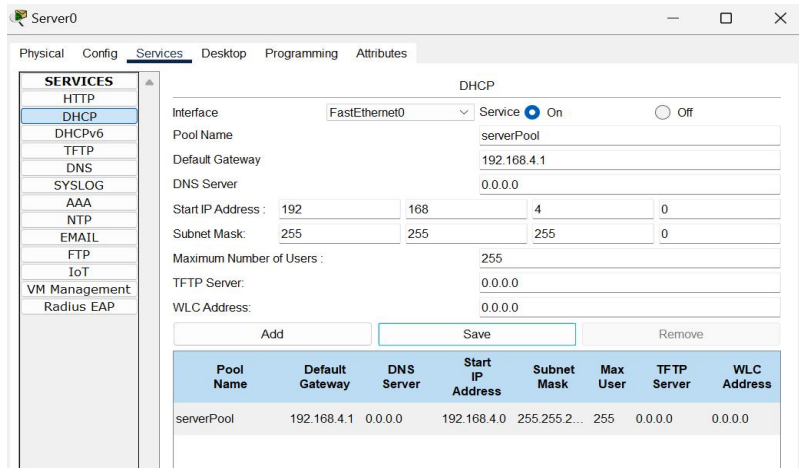
3.2 DHCP Server configuration:



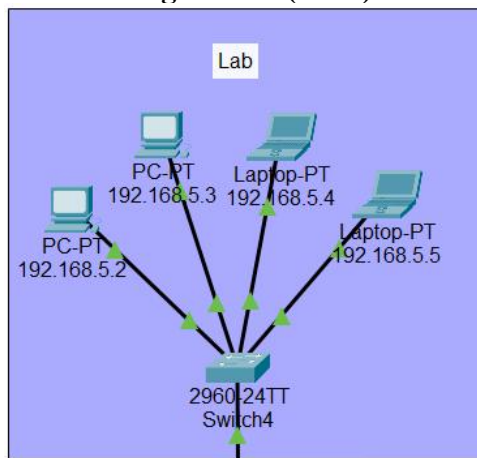
IP Addresses are as follows
192.168.4.2- PC1
192.168.4.3- DHCP server
Subnet Mask- 255.255.255.0
Default Gateway- 192.168.4.1



SS of server services:



3.3 Lab IP configuration (VPN):



IP Addresses are as follows

192.168.5.2- PC1

192.168.5.3- PC2

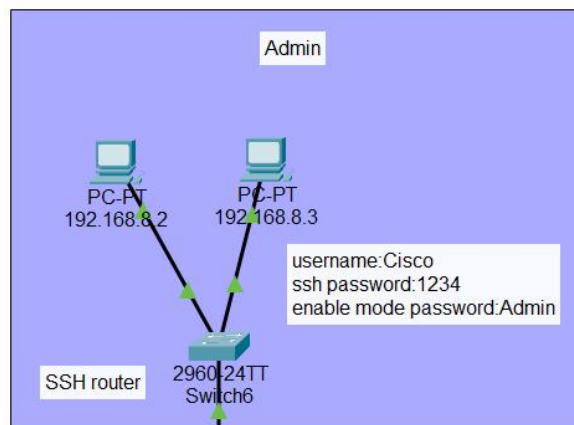
192.168.5.4- PC3

192.168.5.5- PC4

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.5.1

3.4 Admin configuration:



IP Addresses are as follows

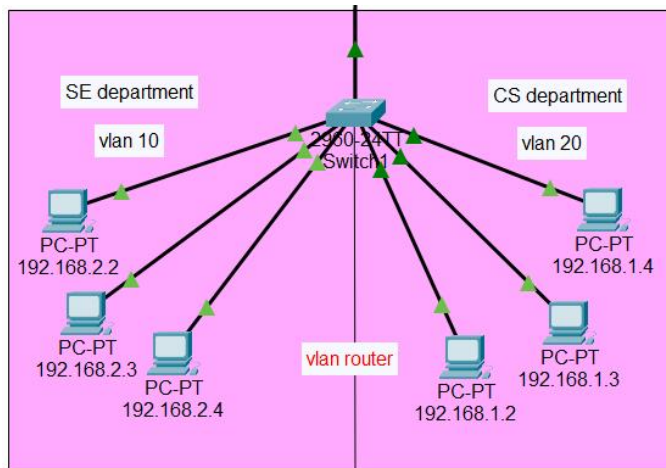
192.168.8.2- PC1

192.168.8.3- PC2

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.8.1

3.5 SE and CS department configuration (vlan 10 and vlan 20)



IP Addresses of **SE department** are as follows

192.168.2.2- PC1

192.168.2.3- PC2

192.168.2.4- PC3

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.2.1

IP Addresses of **CS department** are as follows

192.168.1.2- PC1

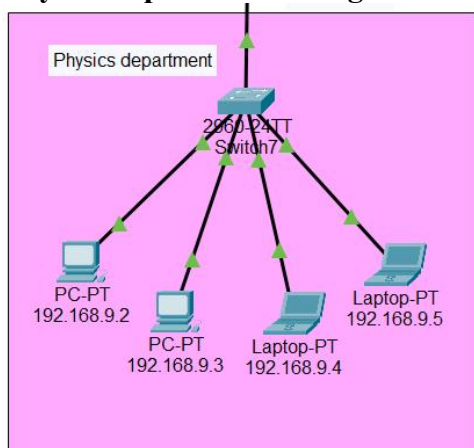
192.168.1.3- PC2

192.168.1.4- PC3

Subnet Mask- 255.255.255.0

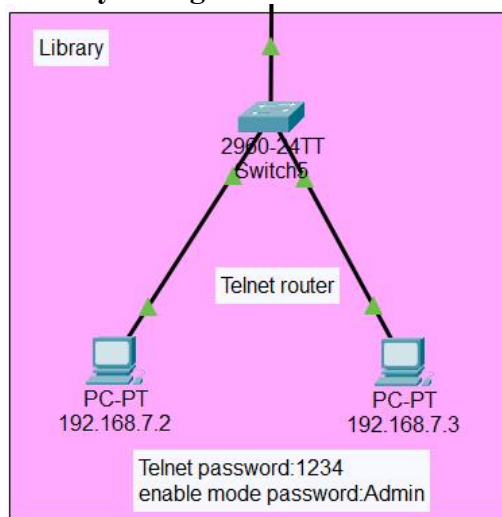
Default Gateway- 192.168.1.1

3.6 Physics department configuration:



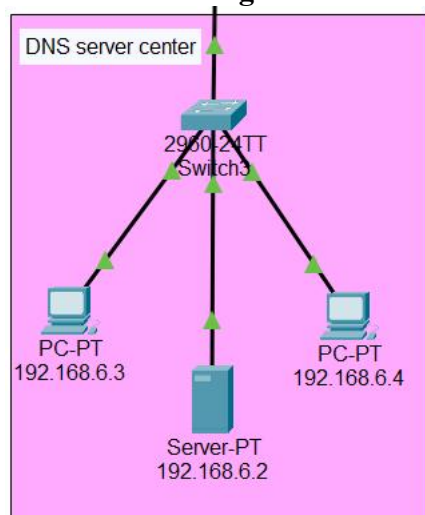
IP Addresses are as follows
192.168.9.2- PC
192.168.9.3- PC
192.168.9.4- Laptop
192.168.9.5- Laptop
Subnet Mask- 255.255.255.0
Default Gateway- 192.168.9.1

3.7 Library configuration:



IP Addresses are as follows
192.168.7.2- PC1
192.168.7.3- PC2
Subnet Mask- 255.255.255.0
Default Gateway- 192.168.7.1

3.8 DNS server configuration:



IP Addresses are as follows
192.168.6.3- PC
192.168.6.4- PC
Subnet Mask- 255.255.255.0

Default Gateway- 192.168.6.1
DNS Server- 192.168.6.2

DNS Services:

No.	Name	Type	Detail
0	www.mycampus.edu	A Record	192.168.6.2

Web Browser

URL: http://www.mycampus.edu

Go Stop

Hello Cisco

4. ROUTERS CONFIGURATIONS AND RIP:

4.1 Router 0 configuration:

Global Settings

Display Name: Router0

Hostname: Router

NVRAM: Erase Save

Startup Config: Load... Export...

Running Config: Export... Merge...

GigabitEthernet0/0:

Port Status: ☒ On

Bandwidth: ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps

Duplex: ☐ Half Duplex ☒ Full Duplex

MAC Address: 0004.9A05.1901

IP Configuration

IPv4 Address: 1.0.0.1

Subnet Mask: 255.0.0.0

GigabitEthernet0/1:

GigabitEthernet0/1	
Port Status	☒ On
Bandwidth	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0004.9A05.1902
IP Configuration	
IPv4 Address	192.168.3.1
Subnet Mask	255.255.255.0

RIP:

Network Address
1.0.0.0
192.168.3.0

4.2 Router 1 configuration:

Router1	
Physical	Config CLI Attributes
GLOBAL	
Settings	
Algorithm Settings	
ROUTING	
Static	
RIP	
SWITCHING	
VLAN Database	
INTERFACE	
GigabitEthernet0/0	

Global Settings	
Display Name	Router1
Hostname	Router
NVRAM	Erase Save
Startup Config	Load... Export...
Running Config	Export... Merge...

GigabitEthernet0/0:

GigabitEthernet0/0	
Port Status	☒ On
Bandwidth	☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0060.708A.2D01
IP Configuration	
IPv4 Address	1.0.0.2
Subnet Mask	255.0.0.0

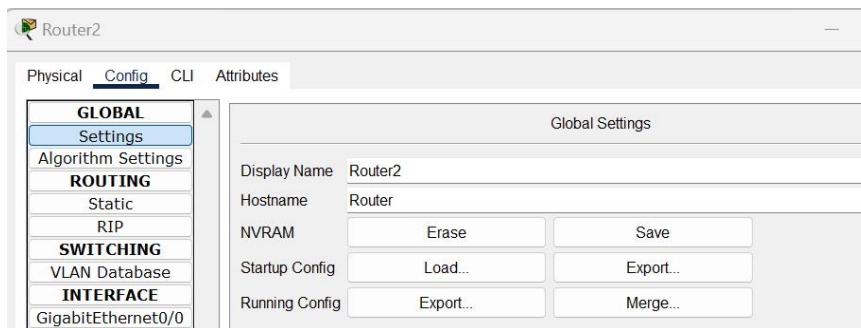
GigabitEthernet0/1:

GigabitEthernet0/1	
Port Status	☒ On
Bandwidth	☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0060.708A.2D02
IP Configuration	
IPv4 Address	2.0.0.1
Subnet Mask	255.0.0.0

RIP:

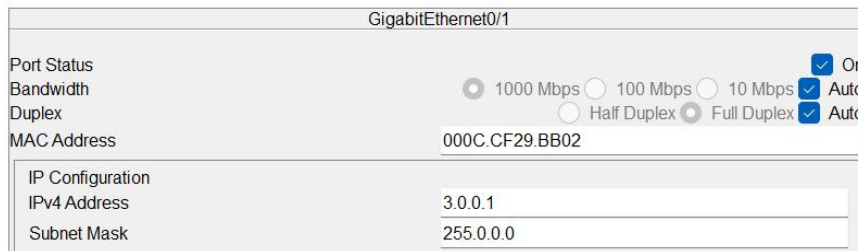
Network Address
1.0.0.0
2.0.0.0
192.168.1.0
192.168.2.0

4.3 Router 2 Configuration:



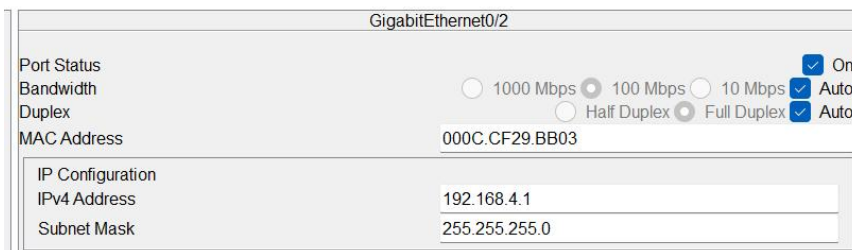
The image shows the configuration interface for Router2. The 'Config' tab is selected, and the 'ROUTING' section is expanded, showing 'RIP' as the active configuration. The 'Global Settings' section is visible, showing the 'Display Name' as 'Router2' and the 'Hostname' as 'Router'. There are buttons for 'Erase', 'Save', 'Load...', 'Export...', and 'Merge...'.

GigabitEthernet0/1:



The image shows the configuration for GigabitEthernet0/1. The 'Port Status' is 'On'. The 'Bandwidth' is set to '1000 Mbps'. The 'Duplex' is set to 'Full Duplex'. The 'MAC Address' is '000C.CF29.BB02'. The 'IP Configuration' section shows the 'IPv4 Address' as '3.0.0.1' and the 'Subnet Mask' as '255.0.0.0'.

GigabitEthernet0/2:

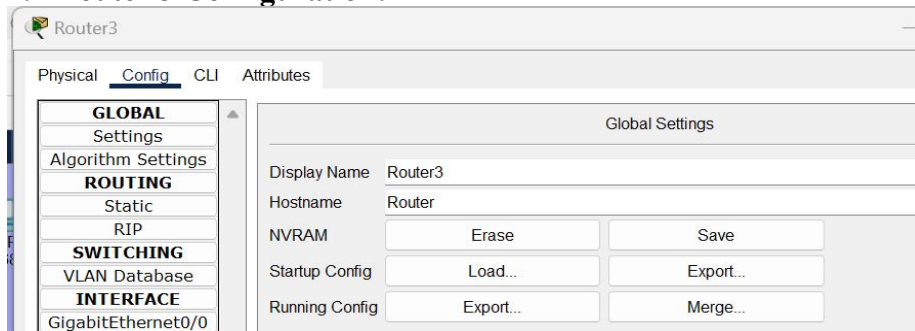


The image shows the configuration for GigabitEthernet0/2. The 'Port Status' is 'On'. The 'Bandwidth' is set to '100 Mbps'. The 'Duplex' is set to 'Full Duplex'. The 'MAC Address' is '000C.CF29.BB03'. The 'IP Configuration' section shows the 'IPv4 Address' as '192.168.4.1' and the 'Subnet Mask' as '255.255.255.0'.

RIP:

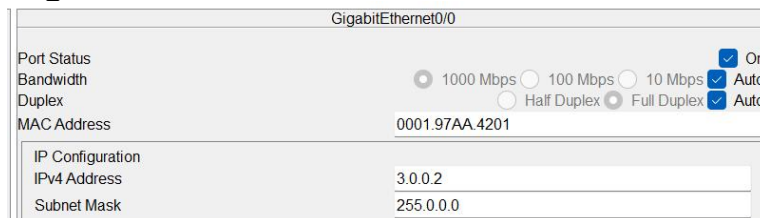
Network Address
2.0.0.0
3.0.0.0
192.168.4.0

4.4 Router 3 Configuration:



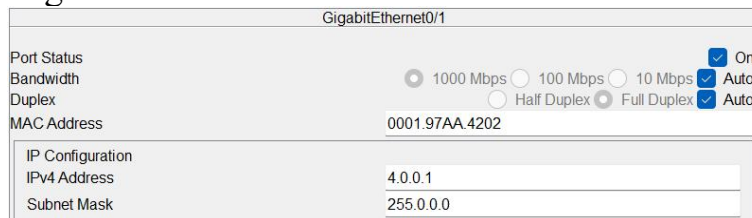
The image shows the configuration interface for Router3. The left sidebar contains a tree view with categories: GLOBAL (Settings, Algorithm Settings), ROUTING (Static, RIP), SWITCHING (VLAN Database), and INTERFACE (GigabitEthernet0/0). The main area is titled 'Global Settings' and contains fields for Display Name (Router3), Hostname (Router), and NVRAM (Erase, Save buttons). Below these are Startup Config (Load..., Export...) and Running Config (Export..., Merge...) buttons.

GigabitEthernet0/0:



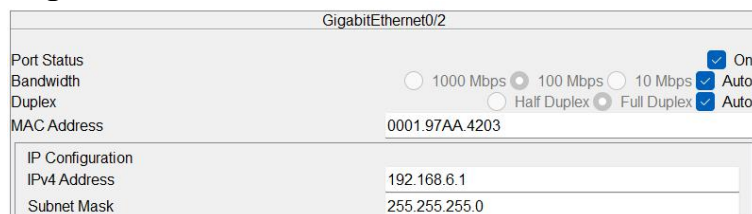
The image shows the configuration for GigabitEthernet0/0. It includes Port Status (On), Bandwidth (1000 Mbps), Duplex (Full Duplex), and MAC Address (0001.97AA.4201). The IP Configuration section shows IPv4 Address (3.0.0.2) and Subnet Mask (255.0.0.0).

GigabitEthernet0/1:



The image shows the configuration for GigabitEthernet0/1. It includes Port Status (On), Bandwidth (100 Mbps), Duplex (Full Duplex), and MAC Address (0001.97AA.4202). The IP Configuration section shows IPv4 Address (4.0.0.1) and Subnet Mask (255.0.0.0).

GigabitEthernet0/2:

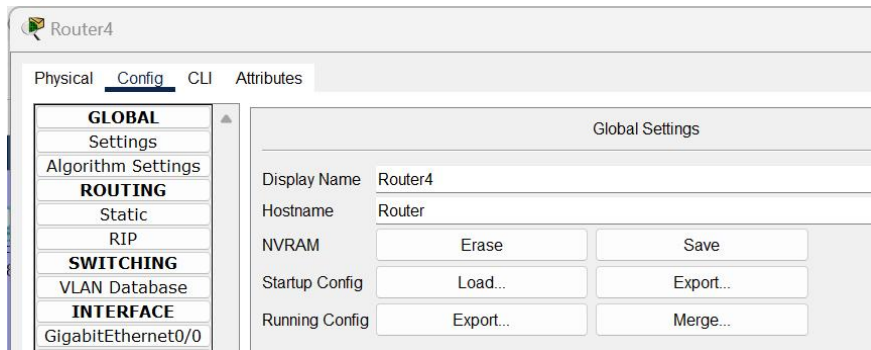


The image shows the configuration for GigabitEthernet0/2. It includes Port Status (On), Bandwidth (100 Mbps), Duplex (Full Duplex), and MAC Address (0001.97AA.4203). The IP Configuration section shows IPv4 Address (192.168.6.1) and Subnet Mask (255.255.255.0).

RIP:

Network Address
3.0.0.0
4.0.0.0
192.168.6.0

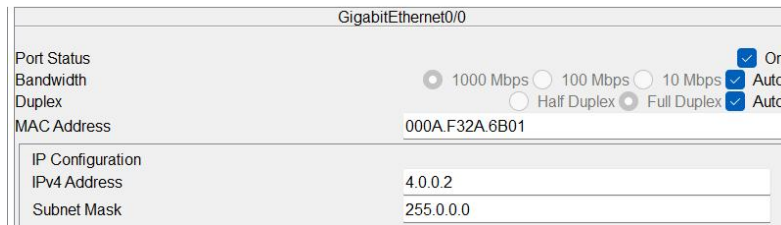
4.5 Router 4 Configuration:



The image shows the configuration interface for Router4. The 'Config' tab is selected, and the 'Global Settings' section is visible. The left sidebar shows a tree view with categories: GLOBAL, ROUTING, SWITCHING, and INTERFACE. Under INTERFACE, 'GigabitEthernet0/0' is selected. The main area displays the following settings:

Global Settings	
Display Name	Router4
Hostname	Router
NVRAM	Erase Save
Startup Config	Load... Export...
Running Config	Export... Merge...

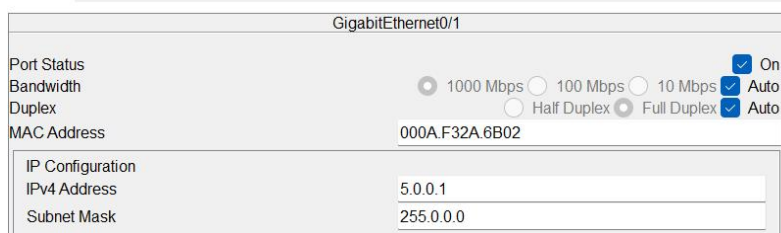
GigabitEthernet0/0:



The image shows the configuration for GigabitEthernet0/0. The settings are as follows:

GigabitEthernet0/0	
Port Status	☒ On
Bandwidth	☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps
Duplex	☐ Half Duplex ☒ Full Duplex
MAC Address	000A.F32A.6B01
IP Configuration	
IPv4 Address	4.0.0.2
Subnet Mask	255.0.0.0

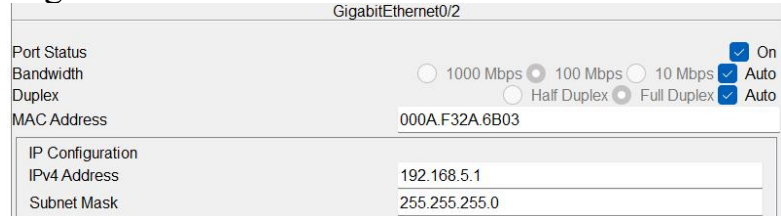
GigabitEthernet0/1:



The image shows the configuration for GigabitEthernet0/1. The settings are as follows:

GigabitEthernet0/1	
Port Status	☒ On
Bandwidth	☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps
Duplex	☐ Half Duplex ☒ Full Duplex
MAC Address	000A.F32A.6B02
IP Configuration	
IPv4 Address	5.0.0.1
Subnet Mask	255.0.0.0

GigabitEthernet0/2:



The image shows the configuration for GigabitEthernet0/2. The settings are as follows:

GigabitEthernet0/2	
Port Status	☒ On
Bandwidth	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps
Duplex	☐ Half Duplex ☒ Full Duplex
MAC Address	000A.F32A.6B03
IP Configuration	
IPv4 Address	192.168.5.1
Subnet Mask	255.255.255.0

RIP:

Network Address
4.0.0.0
5.0.0.0
192.168.5.0

4.6 Router 5 Configuration:

Router5

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

Global Settings

Display Name Router5

Hostname R5

NVRAM

Startup Config

Running Config

GigabitEthernet0/0:

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0006.2A38.E401

IP Configuration

IPv4 Address 5.0.0.2

Subnet Mask 255.0.0.0

GigabitEthernet0/1:

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0006.2A38.E402

IP Configuration

IPv4 Address 6.0.0.1

Subnet Mask 255.0.0.0

GigabitEthernet0/2:

GigabitEthernet0/2

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0006.2A38.E403

IP Configuration

IPv4 Address 192.168.7.1

Subnet Mask 255.255.255.0

RIP:

Network Address
5.0.0.0
6.0.0.0
192.168.7.0

4.7 Router 6 Configuration:

Router6

Physical **Config** CLI Attributes

GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0

Global Settings

Display Name Router6

Hostname R6

NVRAM

Startup Config

Running Config

GigabitEthernet0/0:

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0010.1103.5801

IP Configuration

IPv4 Address 6.0.0.2

Subnet Mask 255.0.0.0

GigabitEthernet0/1:

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0010.1103.5802

IP Configuration

IPv4 Address 192.168.8.1

Subnet Mask 255.255.255.0

GigabitEthernet0/2:

GigabitEthernet0/2

Port Status ☒ On

Bandwidth ☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0010.1103.5803

IP Configuration

IPv4 Address 7.0.0.1

Subnet Mask 255.0.0.0

RIP:

Network Address
6.0.0.0
7.0.0.0
192.168.8.0

4.8 Router 7 Configuration:

The screenshot shows the configuration page for Router7. The left sidebar has a tree view with categories: GLOBAL (Settings, Algorithm Settings), ROUTING (Static, RIP), SWITCHING (VLAN Database), and INTERFACE (GigabitEthernet0/0). The main area is titled 'Global Settings' and contains the following fields and buttons:

- Display Name: Router7
- Hostname: Router
- NVRAM: Erase, Save
- Startup Config: Load..., Export...
- Running Config: Export..., Merge...

GigabitEthernet0/0:

The screenshot shows the configuration page for GigabitEthernet0/0. The left sidebar is partially visible. The main area contains the following settings:

- Port Status: ☒ On
- Bandwidth: ☒ 1000 Mbps, ☐ 100 Mbps, ☐ 10 Mbps
- Duplex: ☐ Half Duplex, ☒ Full Duplex
- MAC Address: 000C.850C.2601
- IP Configuration:
 - IPv4 Address: 7.0.0.2
 - Subnet Mask: 255.0.0.0

GigabitEthernet0/1:

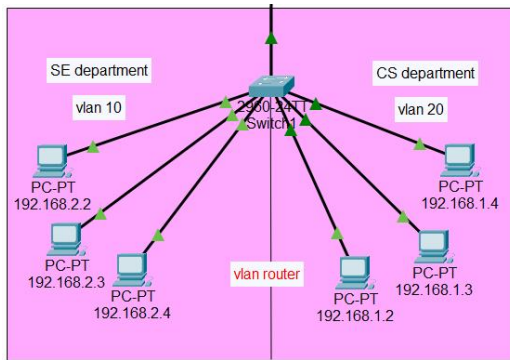
The screenshot shows the configuration page for GigabitEthernet0/1. The left sidebar is partially visible. The main area contains the following settings:

- Port Status: ☒ On
- Bandwidth: ☐ 1000 Mbps, ☒ 100 Mbps, ☐ 10 Mbps
- Duplex: ☐ Half Duplex, ☒ Full Duplex
- MAC Address: 000C.850C.2602
- IP Configuration:
 - IPv4 Address: 192.168.9.1
 - Subnet Mask: 255.255.255.0

RIP:

Network Address
7.0.0.0
192.168.9.0

5. VLAN 10 AND VLAN 20 CONFIGURATION:



Commands for making vlan 10 and vlan 20 for SE and CS department on switch 1:

```
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SE
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name CS
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#interface range fa0/2-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#no shut
Switch(config-if-range)#interface range fa0/5-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
```

Copy

Commands on router 1:

```
Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
Router(config-if)#interface gig0/2.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.10, changed state to up

Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#interface gig0/2.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2.20, changed state to up

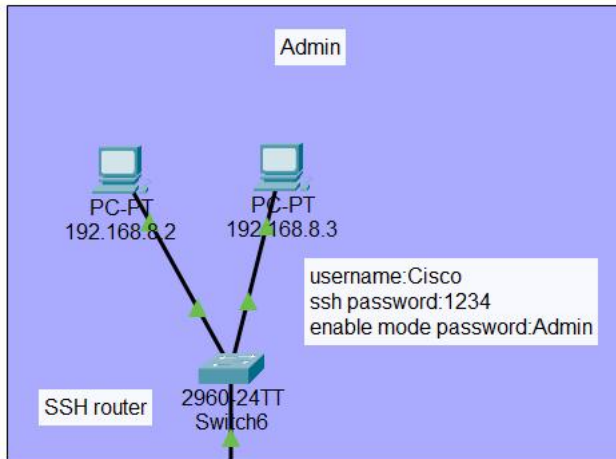
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.20, changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#
```

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6. SSH ROUTING:



Commands on router 6 to achieve SSH:

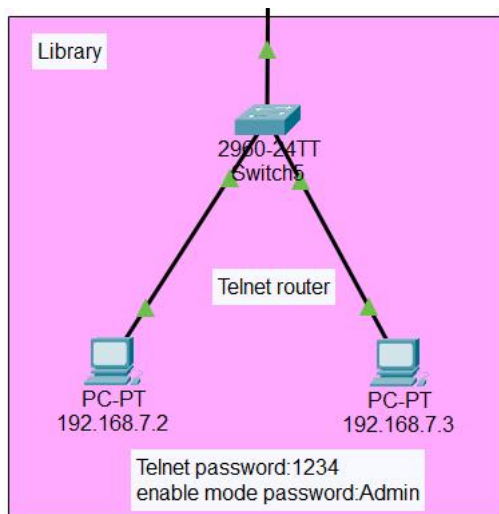
```

Router6
Physical Config CLI Attributes
IOS Command Line Interface
R6>en
R6#config t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#ip domain-name Admin
R6(config)#crypto key generate rsa
% You already have RSA keys defined named R6.Admin .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: R6.Admin
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 800
% Generating 800 bit RSA keys, keys will be non-exportable...[OK]
R6(config)#username Cisco password 1234
*Mar 1 2:19:27.445: %SSH-5-ENABLED: SSH 1.99 has been enabled
R6(config)#ip ssh time-out 30
R6(config)#ip ssh authentication-retries 5
R6(config)#line vty 0 4
R6(config-line)#transport input ssh
R6(config-line)#login local
R6(config-line)#exit
R6(config)#
R6(config)#
R6(config)#
R6(config)#
R6(config)#interface GigabitEthernet0/1
R6(config-if)#ip address 192.168.8.1 255.255.255.0
R6(config-if)#no shutdown
R6(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R6(config-if)#exit
R6(config)#wr
^
% Invalid input detected at '^' marker.
R6(config)#
R6(config)#enable secret Admin
R6(config)#
  
```

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Top

7. TELNET ROUTING:



Commands on router 5 to achieve Telnet routing:

```
Router5
Physical Config CLI Attributes
IOS Command Line Interface

Router>
Router>
Router>
Router>
Router>enable
Router>config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#hostname R5
R5(config)#
R5(config)#int gig0/2
R5(config-if)#ip address 192.168.7.1 255.255.255.0
R5(config-if)#no shutdown

R5(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R5(config-if)#
R5(config-if)#end
R5#
%SYS-5-CONFIG_I: Configured from console by console

R5#wr
Building configuration...
[OK]
R5#
R5#
R5#config t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#enable se
% Incomplete command.
R5(config)#enable secret Admin
R5(config)#end
R5#
%SYS-5-CONFIG_I: Configured from console by console

R5#exit
```

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Top

```

R5>
R5>en
Password:
R5#
R5#config t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#line vty 0 5
R5(config-line)#login
% Login disabled on line 388, until 'password' is set
% Login disabled on line 389, until 'password' is set
% Login disabled on line 390, until 'password' is set
% Login disabled on line 391, until 'password' is set
% Login disabled on line 392, until 'password' is set
% Login disabled on line 393, until 'password' is set
R5(config-line)#password
% Incomplete command.
R5(config-line)#password 1234
R5(config-line)#end
R5#
%SYS-5-CONFIG_I: Configured from console by console

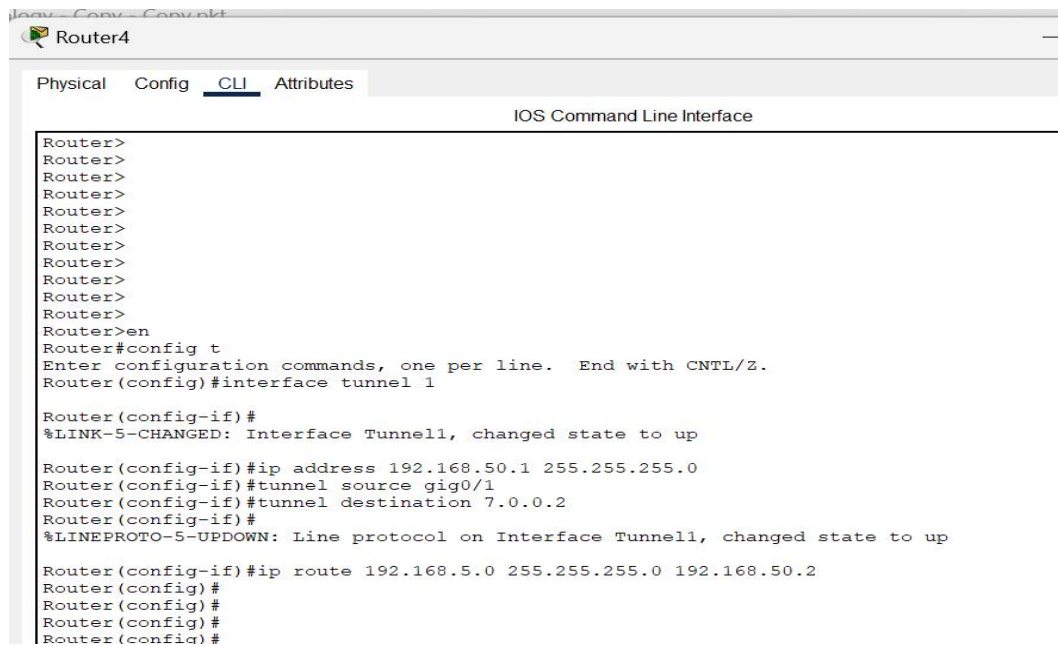
R5#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

R5#exit

```

8. VPN TUNNELLING:

8.1 Between Router 4(lab) and Router 7(physics department): Commands for making VPN tunnel on router 4:



```

Router>
Router>
Router>
Router>
Router>
Router>
Router>
Router>
Router>
Router>
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface tunnel 1

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up

Router(config-if)#ip address 192.168.50.1 255.255.255.0
Router(config-if)#tunnel source gig0/1
Router(config-if)#tunnel destination 7.0.0.2
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up

Router(config-if)#ip route 192.168.5.0 255.255.255.0 192.168.50.2
Router(config)#
Router(config)#
Router(config)#
Router(config)#

```

Commands for making VPN tunnel on router 7:

```

Router7

Physical Config CLI Attributes

IOS Command Line Interface

http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>
Router>
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface tunnel 2

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel2, changed state to up

Router(config-if)#ip address 192.168.50.2 255.255.255.0
Router(config-if)#tunnel source gig0/0
Router(config-if)#tunnel destination 5.0.0.1
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel2, changed state to up

Router(config-if)#ip route 192.168.9.0 255.255.255.0 192.168.50.1
Router(config)#
Router(config)#

```

9. OVERALL STRUCTURE:

